

REGNO:26BCS0019

NAME: R.VAISHNAVI

DSA LAB
LAB TEST

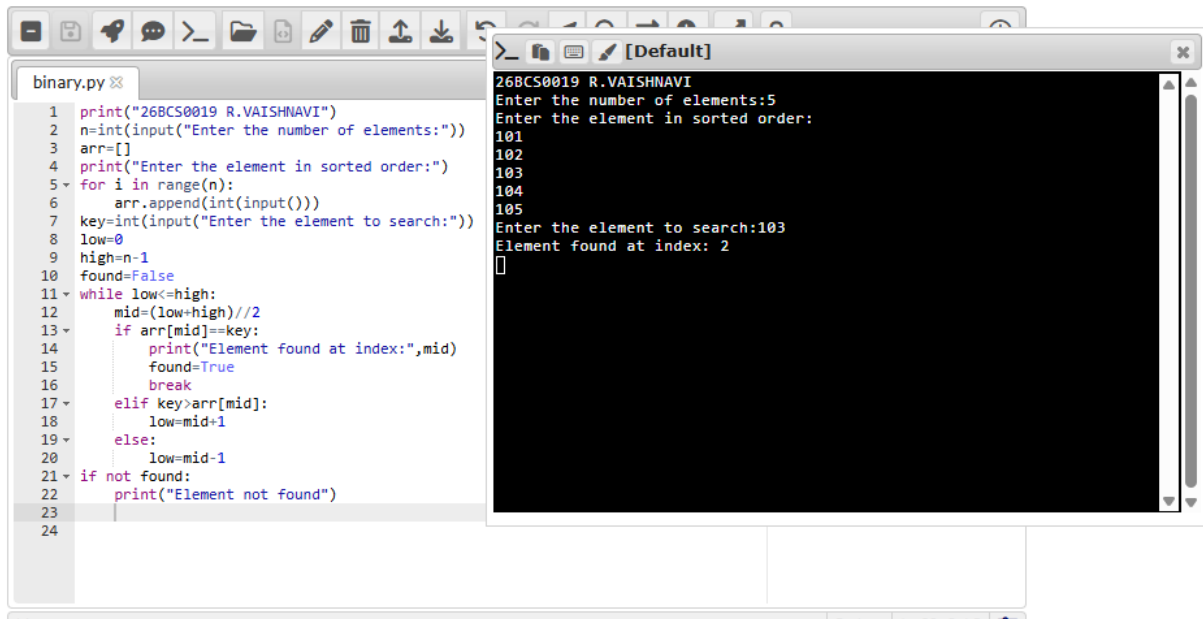
SLOT:L31+L32

DATE:27.07.26

QUESTION SET NO:3

 LAB TEST PROGRAM

 Description  Submission  Edit  Submission view



The screenshot displays a VPL (Virtual Programming Lab) interface. On the left, a code editor window titled 'binary.py' contains a Python program for binary search. The code prompts the user for the number of elements, the elements themselves, and a search key, then performs a binary search and prints the result. On the right, a terminal window titled '[Default]' shows the execution of the program. The user has entered 5 elements: 101, 102, 103, 104, 105, and the search key 103. The program output shows 'Element found at index: 2'.

```
1 print("26BCS0019 R.VAISHNAVI")
2 n=int(input("Enter the number of elements:"))
3 arr=[]
4 print("Enter the element in sorted order:")
5 for i in range(n):
6     arr.append(int(input()))
7 key=int(input("Enter the element to search:"))
8 low=0
9 high=n-1
10 found=False
11 while low<=high:
12     mid=(low+high)//2
13     if arr[mid]==key:
14         print("Element found at index:",mid)
15         found=True
16         break
17     elif key>arr[mid]:
18         low=mid+1
19     else:
20         high=mid-1
21 if not found:
22     print("Element not found")
23
24
```

```
26BCS0019 R.VAISHNAVI
Enter the number of elements:5
Enter the element in sorted order:
101
102
103
104
105
Enter the element to search:103
Element found at index: 2
[]
```